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Hideki Hashimoto, Seiichi Mizuno, Tadashi Nishikawara, Soichi Yoneda & Atsushi Nagazumi

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Randomized trial of carbonated water consumption on snacking behavior, alcohol intake, and health indices in healthy Japanese adults

Hideki Hashimoto ¹⁾ Seiichi Mizuno ¹⁾ Tadashi Nishikawara ²⁾ Soichi Yoneda ³⁾ Atsushi Nagazumi ⁴⁾

¹⁾Asahi Soft Drinks Corporation ²⁾ Macromill, Inc. ³⁾ Mibyou Life Science Corporation ⁴⁾ Kanauchi Medical Clinic

Corresponding author:

Hideki Hashimoto, Ph.D (hideki.hashimoto@asahiinryo.co.jp), ORCID:<https://orcid.org/0009-0001-4019-9988>

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ABSTRACT

Lifestyle-related diseases associated with obesity remain a major public health concern in Japan, in which eating and drinking habits play a critical role. Energy-free carbonated water has been suggested to promote sensations of fullness and may serve as a low-caloric alternative to caloric foods or alcoholic beverages; however, evidence regarding its sustained effects on habitual behaviors and objective health indices is limited. In this randomized, open-label, placebo-controlled, parallel-group trial, 46 healthy Japanese adults (aged 20–64 years; body mass index *BMI*, kg/m², 23.0–<30.0) were allocated to consume either carbonated water (4.3 ± 0.1 g carbon dioxide per 500 g/day) or non-carbonated water once daily for 12 weeks; 45 participants completed the study (carbonated, n = 22; placebo, n = 23). Changes from baseline in snacking frequency, alcohol intake, anthropometric measurements, and blood biochemical parameters were compared between groups. Compared with the placebo group, the carbonated water group showed a greater reduction in snacking frequency (times/week) at week 4 (mean difference −1.39; 95% confidence interval CI −2.82 to −0.28; P = 0.018) and week 12 (−1.49; 95% CI −2.93 to −0.05; P = 0.049). Alcohol intake (g/week) was also reduced at week 12 (−24.44; 95% CI −44.02 to −4.86; P = 0.019). At week 12, modest decreases were observed in body weight and body mass index (*BMI*) compared with placebo (body weight: −0.72 kg; 95% CI −1.37 to −0.07; P = 0.036; *BMI*: −0.26 kg/m²; 95% CI −0.50 to −0.02; P = 0.037). In addition, liver-related enzymes showed favorable changes at week 12, including lower aspartate aminotransferase (*AST*) (−4.05 U/L; 95% CI −7.57 to −0.53; P = 0.034) and alanine aminotransferase (*ALT*) (−9.47 U/L; 95% CI −18.01 to −0.93; P = 0.026). No study beverage-related adverse events were observed. These findings suggest that daily consumption of carbonated water may be associated with favorable changes in snacking and alcohol consumption behaviors and modest improvements in selected anthropometric and liver-related indices. Given the exploratory nature of the analyses and multiple outcomes without multiplicity adjustment, findings should be interpreted cautiously.

KEYWORDS

carbonated water, snacking behavior, alcohol intake, body mass index, liver enzymes

Introduction

The prevalence of lifestyle-related diseases, including obesity, diabetes, hypertension, and dyslipidemia, has increased in Japan due to population aging and westernization of dietary and lifestyle habits. In parallel, the incidence of cardiovascular diseases such as angina pectoris and myocardial infarction has also risen. According to the Vital Statistics for FY2022 published by Ministry of Health, Labour and Welfare (MHLW), heart disease and cerebrovascular disease are the second and fourth leading causes of death in Japan, respectively, together accounting for 21.6% of all deaths [1]. Atherosclerosis is considered a common underlying pathology in many of these conditions, highlighting the importance of lifestyle modification for both prevention and progression of lifestyle-related diseases.

In response to this public health challenge, the Japanese government has promoted nationwide health policies based on the Health Promotion Act. Following the "Second National Health Promotion Movement in the 21st Century (Healthy Japan 21: Phase II)" implemented from 2013 to 2023, the "Third National Health Promotion Movement in the 21st Century (Healthy Japan 21 (Phase III))" was launched in 2024. This initiative continues to emphasize the improvement of dietary habits, alcohol consumption, and physical activity as central strategies for preventing the onset and worsening of lifestyle-related diseases [2].

Despite these efforts, metabolic syndrome remains highly prevalent. According to the FY2022 report on Specific Health Checkups, 16.7% of individuals were classified as having metabolic syndrome, and an additional 12.4% were categorized as having pre-metabolic syndrome. Among adults aged 40–74 years, approximately 15.07 million individuals are estimated to require lifestyle improvement, including 8.65 million with metabolic syndrome and 6.42 million people with pre-metabolic syndrome [3]. Metabolic syndrome is closely associated with obesity-related lipid abnormalities and liver dysfunction [4], indicating that even individuals without overt disease may already exhibit early metabolic and hepatic changes. Therefore, early lifestyle interventions targeting dietary and drinking behaviors are crucial to prevent progression to clinically manifest disease.

Unhealthy lifestyle behaviors, particularly physical inactivity, frequent snacking, excessive caloric intake, and excessive alcohol consumption, are widely recognized as major contributors to metabolic syndrome and related disorders. Accordingly, maintaining a balanced diet and optimizing alcohol consumption are widely recommended as effective preventive strategies [5–7]. However, sustaining long-term behavioral change remains challenging, and simple, low-burden interventions that easily incorporated into daily life are needed.

In recent years, attention has been drawn to the potential role of energy-free carbonated water as a behavioral intervention. Carbonated water is commonly consumed as a non-caloric beverage and has been reported to induce sensations of fullness and refreshment through gastric distension and sensory stimulation. Experimental studies suggest that carbonated water may influence appetite, gastric motility, and sensory-specific satiety, thereby potentially reducing food and alcohol cravings [8, 9]. Nevertheless, evidence regarding whether habitual, long-term consumption of carbonated water leads to meaningful changes in real-world eating and drinking behaviors and objective health parameters remains limited.

Therefore, the present study aimed to investigate the effect of continuous daily consumption of carbonated water on snacking frequency, alcohol intake, and dietary behavior, as well as the associated changes in anthropometric measurements and blood biochemical parameters, in healthy Japanese adult men and women with habitual snacking and drinking behaviors. We hypothesized that consumption of carbonated water would improve eating and drinking habits and consequently lead to favorable changes in body weight and selected metabolic and liver-related biomarkers.

Methods

1. Ethical considerations

This study was approved by the Non-Profit Organization Mibyou Research Ethics Review Committee (IRB number: 16000008, December 6, 2024). The study protocol was prospectively registered in the University Hospital Medical Information Network Clinical Trial Registry (UMIN study ID: UMIN 000053167). All procedures were conducted in accordance with the Declaration of Helsinki (revised in 2013) and the Ethical Guidelines for Life Sciences and Medical Research Involving Human Subjects (partially revised in 2023). Prior to participation, all individuals received a detailed explanation of the study objectives, procedures, potential risks, and benefits. Written informed consent was obtained from all participants before enrollment.

2. Participants

2.1 Recruitment and screening

Participants were recruited from the public through a commercial research panel and participated as paid volunteers. All candidates underwent a screening process, including a medical questionnaire and baseline assessments, prior to enrollment. Individuals who met all inclusion criteria and none of the exclusion criteria were enrolled in the study.

2.2 Inclusion criteria

Participants were required to meet all the following criteria at the time of informed consent.

(1) Japanese men or women aged 20–64 years old; (2) Apparently healthy individuals not receiving any medical treatment for chronic disease; (3) Body mass index (BMI) $\geq 23.0 \text{ kg/m}^2$ and $< 30.0 \text{ kg/m}^2$; (4) Habitual consumption of unsweetened carbonated water $\leq 500 \text{ ml}$ per week; (5) Habitual snacking between meals or at night and consumption sweetened beverages at least twice per week; (6) Habitual alcohol consumption at least once per week; (7) Ability to visit the designated study facility laboratory and complete all required records and questionnaires; (8) Provision of written informed consent after receiving a full explanation of the study;

2.3 Exclusion criteria

Individuals meeting any of the following criteria were excluded:

(1) Presence of diagnosed medical conditions requiring ongoing treatment; (2) Known drug or food allergies; (3) History of hypersensitivity reactions to the test food; (4) Pregnancy and lactation; (5) Blood donation exceeding 200 ml within 1 month or 400 ml within 3 months prior to consent; (6) Habitual alcohol consumption exceeding 60 g of alcohol per day; (7) Habitual smoking of more than 20 cigarettes per day; (8) Regular engagement in high-intensity resistance or strength training (≥ 5 session per week); (9) Inability to refrain from consuming food for specified health use, functional foods, or other health-related supplements during the study period; (10) Inability to consume the test food according to the study protocol; (11) Current participation in other clinical trial, clinical research study, or drug trial; (12) Any other condition judged by the study physician to make the individual unsuitable for participation.

3. Test foods

The test food was commercially available unsweetened carbonated water (product name: Wilkinson Tansan), manufactured by Asahi Soft Drinks Co., Ltd. (Tokyo, Japan). The placebo was non-carbonated water, also manufactured by Asahi Soft Drinks Co., Ltd., and provided in the same polyethylene terephthalate (PET) bottle format. Both the carbonated and non-carbonated waters were commercially available products and were identical in volume, packaging material, and manufacturer, differing only in the presence or absence of carbonation. The carbonated water contained $4.3 \pm 0.1 \text{ g}$ of carbon dioxide per 500g. Participants were instructed to consume 500g of the assigned beverage once daily for 12 weeks. All beverages were stored at room temperature under light-shielded conditions in accordance with the manufacturer's recommendations. The best-before date of the beverages used in this study was

July 8, 2024, ensuring product quality and stability throughout the intervention period. Beverages were distributed to participants within the designated.

4. Study design

The study was designed as a randomized, open-label, placebo-controlled, parallel-group trial to evaluate the effect of continuous consumption of carbonated water on eating and drinking behaviors and health examination indices. A total of 46 eligible participants were randomly allocated in a 1:1 ratio to either the test food group (carbonated water) or the placebo groups (non-carbonated water). Randomization was performed using a computer-generated stratified block randomization method, with by age, sex, and body mass index (BMI) at baseline to ensure balance between groups. Blinding of participants was not feasible because carbonation produces distinct sensory characteristics, including effervescence and oral stimulation, which are readily perceptible. Therefore, an open-label design was adopted. To minimize potential bias associated with the lack of blinding, objective outcome measures such as anthropometric measurements and biochemical parameters were included as key endpoints. Participants in both groups were instructed to consume 500 g of the assigned beverage once daily for 12 weeks. They were asked to maintain their usual diet, physical activity, and lifestyle habits throughout the study period. No specific dietary or exercise interventions were imposed. However, participants were instructed to refrain from consuming unsweetened carbonated water other than the test food during the study period. The primary outcome was the between-group difference in changes from baseline in snacking frequency and alcohol intake, assessed using dietary journals. Secondary outcomes included anthropometric measures (body weight, BMI, and blood pressure) and blood biochemical parameters.

5. Items of assessment

(1) Values of dietary journal analysis

The participants were asked to record all food and beverage consumption for 1 week in their dietary journals at baseline and 4, 8, and 12 weeks from the start of consumption of the test foods (baseline). Dietary journals were evaluated for calorific value (energy) and the three macronutrients (proteins, carbohydrates, and lipids) at each interval by a registered dietitian using the Nutrient Calculator Healthy Maker Pro 501 (Mushroomsoft, Co., Ltd.).

(2) Snacking and drinking habits

Daily questionnaires were administered on the frequency of snacking and alcohol consumption from 2 weeks before baseline until 12 weeks after the start of consumption. The survey was completed online in real time using a web-based lifestyle journal. Snacking frequency was defined as the number of times some food was consumed in addition to breakfast, lunch, and dinner. The amount of alcohol consumed was converted into pure alcohol equivalent, and the average value per week during each period was used to evaluate parameters at various intervals.

(3) Physical measurements

Physical measurements were conducted at baseline and 4, 8, and 12 weeks after the start of the intervention. All measurements were performed at the designated study facility trained staff using standardized procedures. Body weight was measured using a calibrated digital scale, with participants wearing light indoor clothing and no shoes. Height was measured at baseline using a stadiometer, and body mass index (BMI) was calculated as body weight (kg) divided by height squared (m^2). Blood pressure was measured in the seated position after a rest period of at least 5 minutes. Measurements were taken on the upper arm using an automated sphygmomanometer. Two consecutive readings were obtained at each time point, and the average of the two measurements was used for analysis. All physical measurements were conducted at approximately the same time of day for each participant across

study visits whenever possible. Participants were instructed to refrain from vigorous physical activity and alcohol consumption on the day prior to measurement.

(4) Blood test

Blood samples were collected at baseline and at 4, 8, and 12 weeks after the start of the intervention. Participants were instructed to fast for at least 8 hours prior to blood collection and to abstain from alcohol consumption on the day before sampling. Venous blood samples were obtained by qualified medical laboratory (SRL, Inc., Tokyo, Japan) and were performed using standardized laboratory procedures and quality-controlled automated clinical chemistry analyzers in accordance with routine clinical testing protocols. The following blood parameters were assessed at each time point: triglycerides, total cholesterol, low-density lipoprotein cholesterol, high-density lipoprotein cholesterol, phospholipids, fasting blood glucose, glycated hemoglobin (NGSP), aspartate transaminase (AST), alanine transaminase (ALT), gamma-glutamyl transferase, alkaline phosphatase, lactate dehydrogenase, total bilirubin, total protein, albumin, uric acid, urea nitrogen, creatinine, sodium, chlorine, potassium, calcium, magnesium, and serum iron. Serum AST and ALT were measured using Sikarikit AST and Sikarikit ALT reagents (Kanto Chemical Co., Inc, Japan) on an automated analyzer (JCA-BM8000 series; JEOL Ltd., Japan) according to the JSCC standardized method.

(5) Safety evaluation

Adverse events and side effects that occurred from baseline to 12 weeks after consumption were tabulated for the individuals who had consumed the test food at least once. Adverse events were defined as all undesirable or unintended injuries and diseases, or signs of them, that occurred to the individual, whether or not causally related to the test food or the tests performed in the study. Among adverse events, side effects were defined as those for which the study investigator determined that a causal relationship with the test food could not be completely ruled out.

6. Statistical analysis

Statistical analyses were performed using IBM SPSS Statistics (ver 29.0.2.0; IBM Corp., Armonk, NY, USA). Continuous variables are presented as mean $\pm$ standard deviation (SD). All statistical tests were two-tailed, and the significance level was set at $P < 0.05$. Changes from baseline at each time point (4, 8, 12 weeks) were calculated for each outcome variable. Between-group comparisons of changes from baseline were conducted using an independent two-sample t-test, while within-group comparisons relative to baseline were assessed using a one-sample t-test. For each outcome, the between-group mean difference in change from baseline was reported together with the exact two-sided P value and the corresponding 95% confidence interval (CI). CIs were calculated using the t distribution based on the standard error of the difference between group mean changes. As this was an exploratory study with multiple outcomes and no adjustment for multiplicity, the results should be interpreted cautiously. No formal sample size calculation was performed prior to study initiation. As a supplementary descriptive analysis, standardized effect sizes (Cohen's d) were calculated post hoc for the primary outcomes (snacking frequency and alcohol intake) to aid interpretation of the magnitude of between-group differences. These effect size estimates were not used for hypothesis testing and should be interpreted cautiously.

Results

1. Participants and groups included in the analysis

The flow from participant inclusion to analysis is shown in Figure 1. Of the 46 individuals included, 45 completed the study; one dropped out during the study period for reasons unrelated to the study. The 22 participants in the test food group and 23 in the placebo group who completed the study were thus included in the analysis.

2. Values of dietary journal analysis

Table 1 shows the analysis of the dietary journals. In terms of changes in values of the dietary journal analysis, the carbonated water group showed a slight decrease ($P = 0.096$) in carbohydrate consumption compared to the non-carbonated water group at 12 weeks, while the analysis of snacks only showed a slight decrease ($P = 0.067$) at 12 weeks. However, there were no significant differences between the test food groups for all items.

3. Snacking and drinking habits

Aggregate results for frequency of snacks and amount of alcohol consumed are shown in Table 2. Compared with the non-carbonated water group, the carbonated water group showed a greater reduction in snacking frequency at week 4 (between-group mean difference in change: -1.39 times/week; 95% CI -2.82 to -0.28 ; $P = 0.018$), corresponding to a moderate-to-large standardized effect size (Cohen's $d = 0.73$). A similar effect was observed at week 12 (mean difference -1.49 ; 95% CI -2.93 to -0.05 ; $P = 0.049$; Cohen's $d = 0.61$). Alcohol intake was also significantly reduced at week 12 in the carbonated water group (mean difference -24.44 g/week; 95% CI -44.02 to -4.86 ; $P = 0.019$), with a moderate standardized effect size (Cohen's $d = 0.73$).

4. Physical measurements

The results of the physical measurements are shown in Table 3. At week 12, body weight and BMI decreased modestly in the carbonated water group compared with the non-carbonated water group (body weight: -0.72 kg; 95% CI -1.37 to -0.07 ; $P = 0.036$; BMI: -0.26 kg/m²; 95% CI -0.50 to -0.02 ; $P = 0.037$).

5. Blood test

The results of the blood test parameters are shown in Table 4. At week 12, liver enzymes decreased in the carbonated water group compared with placebo, including AST (-4.05 U/L; 95% CI -7.57 to -0.53 ; $P = 0.034$) and ALT (-9.47 U/L; 95% CI -18.01 to -0.93 ; $P = 0.026$). Triglycerides showed a trend toward reduction at weeks 8 and 12 ($P = 0.083$ and 0.074 , respectively).

6. Safety assessment

During the study period, adverse events were identified in six individuals in the test food group (26.1% incidence) and eight individuals in the placebo group (34.8% incidence). The adverse events identified were transient or seasonal, such as menstrual cramps, headaches, cold symptoms, and hay fever, and were judged by the study investigators to have no causal relationship to the study foods, suggesting good safety of the test food.

Discussion

This randomized, open-label, placebo-controlled, parallel-group study investigated the effects of 12 weeks of continuous consumption of carbonated water on eating and drinking behaviors and selected health examination indices in healthy Japanese adults with habitual snacking and drinking behaviors. The primary finding of this exploratory study was a reduction in snacking frequency in the carbonated water group compared with the placebo group. In addition, modest changes were

observed in alcohol consumption, body weight, body mass index, and selected blood biochemical parameters, including liver-related enzymes.

The observed reduction in snacking frequency suggests that habitual consumption of carbonated water may support changes in eating behavior. Previous experimental studies have shown that ingestion of carbonated water can induce gastric distension due to both liquid volume and carbon dioxide release, which may enhance sensations of fullness and suppress appetite through reduced gastric motility [8, 9]. Although the present study did not directly assess appetite, gastric motility, or satiety-related hormones, the sustained decrease in snacking frequency observed over the intervention period is consistent with these proposed physiological mechanisms. In addition, the refreshing sensory properties of carbonated water may contribute to temporary stress relief or substitution for other caloric beverages, which could indirectly influence eating and drinking behaviors.

In this study, reductions in alcohol consumption were also observed in the carbonated water group at 12 weeks. One possible explanation is that carbonated water served as a non-caloric alternative beverage, reducing opportunities for alcohol intake. Behavioral substitution is a commonly employed strategy in lifestyle interventions, and the ease of incorporating carbonated water into daily routines may represent a practical advantage. However, because alcohol intake was self-reported, these findings should be interpreted cautiously.

Modest decreases in body weight and BMI were observed in the carbonated water group compared with the placebo group at 12 weeks. These changes are likely secondary to alterations in eating and drinking behaviors rather than a direct metabolic effect of carbonated water itself. Weight reduction associated with improvements in dietary habits has been well documented [10], and even small reductions in body weight may contribute to favorable metabolic changes. Consistent with this interpretation, changes in triglycerides and liver-related enzymes (AST and ALT) were modest and remained within clinically normal ranges. These findings suggest potential improvements in metabolic and hepatic load rather than clinically therapeutic effects.

Several limitations of this study should be acknowledged. First, the study employed an open-label design because blinding was not feasible due to the perceptible sensory characteristics of carbonation. Although objective outcomes such as body weight and blood biochemical parameters were included, the lack of blinding may have influenced self-reported behavioral outcomes. Second, snacking frequency and alcohol intake were assessed using self-reported questionnaires, which are subject to recall and reporting bias. Third, physical activity levels were not quantitatively assessed, and although participants were instructed to maintain their usual activity patterns, unmeasured changes in physical activity may have influenced the results.

In addition, this study was exploratory in nature and was not powered to detect small effect sizes. No formal sample size calculation was performed prior to study initiation, and multiple outcomes were analyzed without adjustment for multiplicity. Therefore, findings showing marginal statistical significance should be interpreted with caution. Furthermore, the study population consisted of healthy Japanese adults with specific BMI ranges and habitual snacking and drinking behaviors, which limits the generalizability of the findings to other populations, including individuals with diagnosed metabolic diseases.

Despite these limitations, this study has several strengths. It evaluated the effects of a commercially available, energy-free beverage that can be easily incorporated into daily life, enhancing the real-world relevance of the findings. The inclusion of both behavioral outcomes and objective health examination indices provides complementary perspectives on lifestyle-related changes. Taken together, the results suggest that continuous consumption of carbonated water may support healthier eating and drinking behaviors and contribute to modest improvements in selected health indicators in healthy adults.

Further studies with larger sample sizes, longer follow-up periods, objective assessments of physical activity and dietary intake, and more rigorous control of confounding factors are warranted to confirm these findings and to clarify the mechanisms underlying the observed behavioral changes.

Conclusion

This exploratory randomized study suggests that continuous consumption of carbonated water may support favorable changes in eating and drinking behaviors, particularly reducing snacking frequency, in healthy adults with habitual snacking and drinking behaviors. Modest improvements in body weight, body mass index, and selected blood biochemical parameters were also observed; however, these changes are likely secondary to behavioral modifications rather than direct physiological effects of carbonated water. Because this study employed an open-label design, relied partly on self-reported behavioral data, and was conducted with a limited sample size without formal power calculation, the findings should be interpreted with caution. The results do not establish causality but indicate a potential role for carbonated water as an easy implementable, energy-free beverage within daily life. Further well-powered, long-term studies incorporating objective assessments of dietary intake, physical activity, and appetite-related mechanisms are needed to confirm these findings and clarify the underlying pathways through which carbonated water consumption may influence eating and drinking behaviors.

Statements and declarations

Funding

This study was funded by Asahi Soft Drinks Co., Ltd.

Conflicts of interest

The research expenses and test foods for this study were provided by Asahi Soft Drinks Co., Ltd. Third-party organization Mibyou Life Science Co. was commissioned to conduct this study under appropriate management on registered subjects of Macromill, Inc. As for the authors, Hideki Hashimoto and Seiichi Mizuno are employees of Asahi Soft Drinks Co., Ltd, Soichi Yoneda is an employee of Mibyou Life Science Co., and Atsushi Nagazumi is the director of the medical corporation Ryoshukai Kanauchi Medical Clinic and was commissioned by Mibyou Life Science Co. to conduct this study as the investigator. Tadashi Nishikawara is an employee of Macromill, Inc. and was commissioned by Asahi Soft Drinks Co., Ltd. to recruit subjects.

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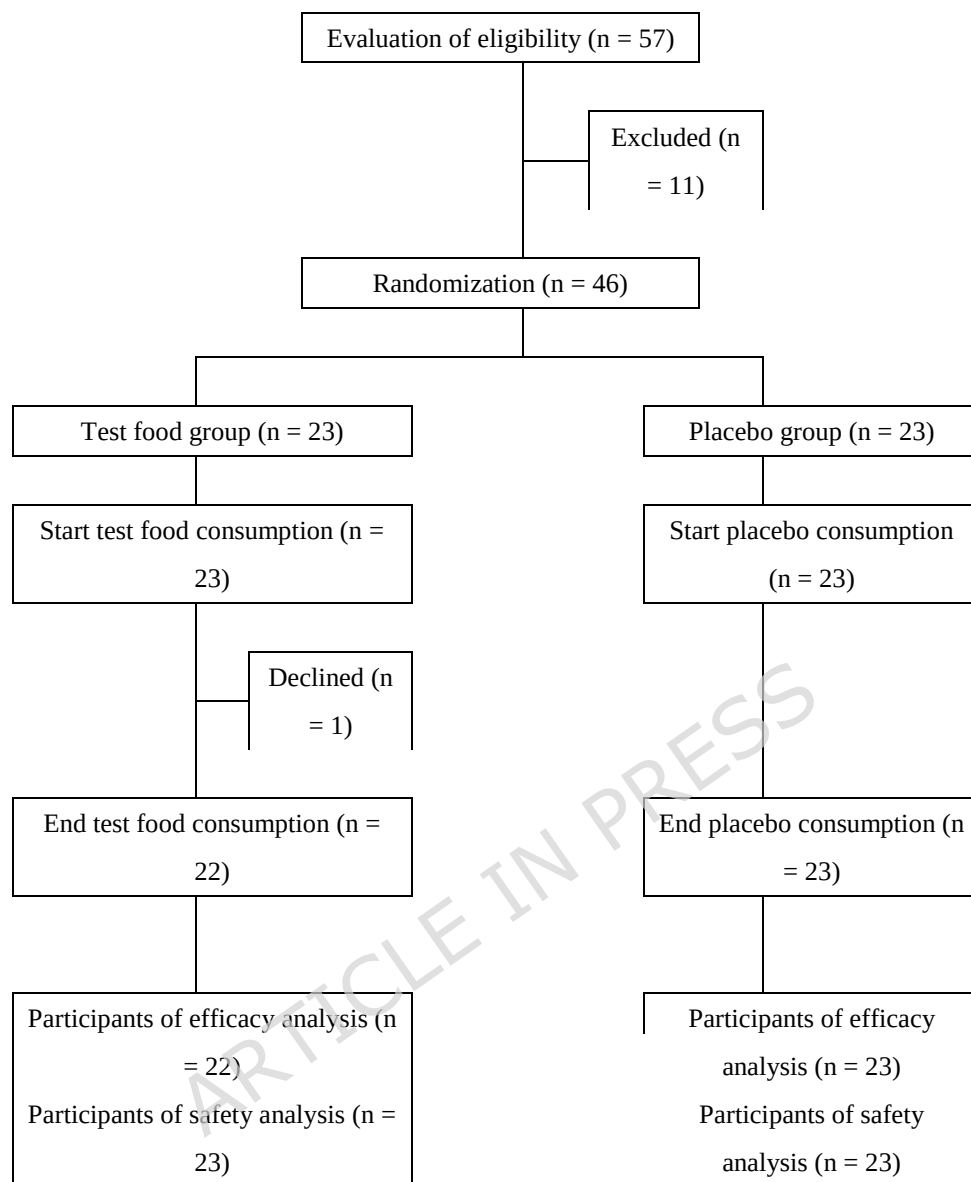
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Figure caption**Figure 1:** Participant follow-up flowchart**Figure 1**

Tables

Table 1. Values of dietary journal analysis

Items		Weeks	Measured values				Change from baseline									
(Units)			Carbonated water group		Non-carbonated water group		Carbonated water group		Non-carbonated water group		Intergroup comparison					
											Mean difference	95%CI		P-value		
Calorific value (kcal/week)	Baseline		13312.2 ±	2053.8	12493.4 ±	2470.1	—		—		—	—	—	—		
	4 wks.		13378.1 ±	2621.3	12466.6 ±	2639.3	65.9 ±	2162.1	-26.9 ±	1722.8	92.8	-1088.21	to	1273.81	0.874	
	8 wks.		12636.8 ±	1776	12054 ±	2462.9	-675.5 ±	1534.7	-439.5 ±	1781.4	-236.0	-1234.58	to	762.58	0.637	
	12 wks.		12450.3 ±	1680.8*	12307.6 ±	2347.9	-861.9 ±	1775.6	-185.8 ±	1931.4	-676.1	-1790.80	to	438.60	0.229	
Protein (g/week)	Baseline		402.6 ±	77.8	387.4 ±	79.5	—		—		—	—	—	—		
	4 wks.		411.2 ±	97.7	383.4 ±	72.2	8.6 ±	88.8	-4.0 ±	64.1	12.60	-34.31	to	59.51	0.586	
	8 wks.		393.8 ±	62.9	374.2 ±	79.9	-8.8 ±	69.9	-13.2 ±	72.7	4.40	-38.47	to	47.27	0.835	
	12 wks.		385.1 ±	63.1	385.0 ±	92.1	-17.5 ±	81.2	-2.3 ±	82.6	-15.20	-64.45	to	34.05	0.539	
Carbohydrates (g/week)	Baseline		1864.6 ±	377.8	1806.9 ±	380.5	—		—		—	—	—	—		
	4 wks.		1826.0 ±	389.7	1761.5 ±	366	-38.6 ±	271.1	-45.4 ±	227.6	6.80	-144.23	to	157.83	0.927	
	8 wks.		1719.1 ±	310.0*	1664.2 ±	355.2*	-145.5 ±	231.7	-142.7 ±	261.6	-2.80	-151.23	to	145.63	0.970	
	12 wks.		1669.6 ±	332.1*	1740.1 ±	303.8	-195.1 ±	258	-66.8 ±	248.1	-128.3	-280.61	to	24.01	0.096	
Lipids (g/week)	Baseline		431.3 ±	81.4	382.7 ±	117.0	—		—		—	—	—	—		
	4 wks.		444.9 ±	131.6	389.6 ±	105.9	13.6 ±	129.6	6.8 ±	69.5	6.80	-56.77	to	70.37	0.830	
	8 wks.		422.1 ±	71.7	396.3 ±	117.5	-9.2 ±	77.6	13.6 ±	93.1	-22.80	-74.27	to	28.67	0.378	
	12 wks.		431.0 ±	90.2	388.9 ±	109.3	-0.3 ±	111.4	6.2 ±	102.3	-6.50	-70.91	to	57.91	0.839	
Snack only: calorific value (kcal/week)	Baseline		1708.3 ±	1153.5	1617 ±	1155.0	—		—		—	—	—	—		
	4 wks.		2076.8 ±	1734.7	1745 ±	1246.0	368.5 ±	1090.4	128.0 ±	858.3	240.5	-352.50	to	833.50	0.414	
	8 wks.		1538.2 ±	1317.7	1456.6 ±	1046.2	-170.1 ±	764.8	-160.4 ±	690.5	-9.70	-448.66	to	429.26	0.964	
	12 wks.		1434.3 ±	1136.9	1597.7 ±	907.8	-274.0 ±	867.9	-19.3 ±	666.3	-254.70	-722.58	to	213.18	0.274	
Snacks only: Protein (g/week)	Baseline		29.9 ±	25.9	28.4 ±	21.6	—		—		—	—	—	—		
	4 wks.		35.4 ±	28.9	35.7 ±	28.4	5.5 ±	16.3	7.3 ±	20.1	-1.80	-12.79	to	9.19	0.733	
	8 wks.		24.0 ±	20.3	23.9 ±	19.1	-5.9 ±	17.3	-4.5 ±	16.8	-1.40	-11.66	to	8.86	0.782	
	12 wks.		23.3 ±	20.7	25.4 ±	16.7	-6.7 ±	17.5	-3.1 ±	14.4	-3.60	-13.27	to	6.07	0.453	
Snacks only: Carbohydrates (g/week)	Baseline		245.2 ±	159.7	236.6 ±	191.5	—		—		—	—	—	—		
	4 wks.		284.3 ±	259.7	237.6 ±	183.5	39.1 ±	158.6	1.0 ±	111	38.1	-44.89	to	121.09	0.354	
	8 wks.		198.0 ±	175.4	207.9 ±	187.4	-47.1 ±	122.7	-28.6 ±	114.3	-18.5	-89.90	to	52.9	0.603	
	12 wks.		179.9 ±	149.7*	232.9 ±	166.1	-65.3 ±	124.3	-3.7 ±	93.9	-61.6	-128.24	to	5.04	0.067	
Snack only: Lipids (g/week)	Baseline		57.5 ±	50.8	51.1 ±	40.4	—		—		—	—	—	—		
	4 wks.		75.5 ±	70.5	58.5 ±	44.3	18.0 ±	51.0	7.4 ±	34.2	10.60	-15.76	to	36.96	0.416	
	8 wks.		54.8 ±	65.5	47.0 ±	32.9	-2.7 ±	35.7	-4.1 ±	27.6	1.40	-17.89	to	20.69	0.885	
	12 wks.		58.6 ±	55.1	52.0 ±	33.7	1.1 ±	34.9	0.9 ±	33.6	0.20	-20.41	to	20.81	0.984	
Values are mean ± standard deviation																
*P < 0.05 vs. baseline within group																
Mean difference = change from baseline (Test food group - Placebo group)																
95% CI calculated using Welch's t method (n = 22 vs. 23)																

Table 2: Snacking and drinking habits

Items		Weeks	Measured values						Change from baseline								
			Carbonated water group			Non-carbonated water group			Carbonated water group			Non-carbonated water group			Intergroup comparison		
(Units)															Mean difference	95% CI	P-value
Body weight (kg)		Baseline	68.57	±	9.10	68.83	±	9.62	—			—			—	—	—
		4 wks.	69.30	±	9.07*	69.10	±	9.63	0.73	±	1.08	0.28	±	1.23	0.45	-0.22 to 1.12	0.201
		8 wks.	68.85	±	9.01	69.22	±	9.35	0.28	±	1.15	0.40	±	1.12	-0.12	-0.80 to 0.56	0.728
		12 wks.	68.45	±	9.27	69.42	±	9.44*	-0.12	±	1.25	0.60	±	0.96	-0.72	-1.37 to -0.07	0.036
Body mass index (kg/m ²)		Baseline	25.26	±	1.49	25.33	±	1.38	—			—			—	—	—
		4 wks.	25.55	±	1.52*	25.43	±	1.32	0.29	±	0.42	0.10	±	0.43	0.19	-0.05 to 0.43	0.134
		8 wks.	25.39	±	1.57	25.48	±	1.28	0.12	±	0.43	0.15	±	0.38	-0.03	-0.26 to 0.20	0.836
		12 wks.	25.22	±	1.63	25.55	±	1.32*	-0.04	±	0.46	0.22	±	0.36	-0.26	-0.50 to -0.02	0.037
Systolic blood pressure (mmHg)		Baseline	121.68	±	10.65	127.65	±	12.12	—			—			—	—	—
		4 wks.	121.59	±	10.34	127.74	±	9.79	-0.09	±	9.84	0.09	±	11.87	-0.18	-6.73 to 6.37	0.957
		8 wks.	119.95	±	9.70	126.35	±	11.88	-1.73	±	8.57	-1.30	±	11.63	-0.43	-6.56 to 5.70	0.891
		12 wks.	120.50	±	11.63	125.17	±	14.16	-1.18	±	11.67	-2.48	±	11.30	1.30	-5.61 to 8.21	0.707
Diastolic blood pressure (mmHg)		Baseline	81.41	±	8.97	84.65	±	10.10	—			—			—	—	—
		4 wks.	81.82	±	8.42	82.96	±	10.39	0.41	±	9.28	-1.70	±	7.23	2.11	-2.92 to 7.14	0.399
		8 wks.	79.23	±	10.9	81.3	±	10.92	-2.18	±	7.61	-3.35	±	8.44	1.17	-3.66 to 6.00	0.629
		12 wks.	78.77	±	10.75	81.96	±	9.21	-2.64	±	6.50	-2.70	±	8.00	0.06	-4.32 to 4.44	0.978
Values are mean ± standard deviation																	
*P < 0.05 vs. baseline within group																	
Mean difference = change from baseline (Test food group - Placebo group)																	
95% CI calculated using Welch's t method (n = 22 vs. 23)																	

Table 3. Results of physical measurements

Items		Weeks	Measured values						Change from baseline								
			Carbonated water group			Non-carbonated water group			Carbonated water group			Non-carbonated water group			Intergroup comparison		
(Units)															Mean difference	95% CI	P-value
Body weight (kg)		Baseline	68.57	±	9.10	68.83	±	9.62	—			—			—	—	—
		4 wks.	69.30	±	9.07*	69.10	±	9.63	0.73	±	1.08	0.28	±	1.23	0.45	-0.22 to 1.12	0.201
		8 wks.	68.85	±	9.01	69.22	±	9.35	0.28	±	1.15	0.40	±	1.12	-0.12	-0.80 to 0.56	0.728
		12 wks.	68.45	±	9.27	69.42	±	9.44*	-0.12	±	1.25	0.60	±	0.96	-0.72	-1.37 to -0.07	0.036
Body mass index (kg/m ²)		Baseline	25.26	±	1.49	25.33	±	1.38	—			—			—	—	—
		4 wks.	25.55	±	1.52*	25.43	±	1.32	0.29	±	0.42	0.10	±	0.43	0.19	-0.05 to 0.43	0.134
		8 wks.	25.39	±	1.57	25.48	±	1.28	0.12	±	0.43	0.15	±	0.38	-0.03	-0.26 to 0.20	0.836
		12 wks.	25.22	±	1.63	25.55	±	1.32*	-0.04	±	0.46	0.22	±	0.36	-0.26	-0.50 to -0.02	0.037
Systolic blood pressure (mmHg)		Baseline	121.68	±	10.65	127.65	±	12.12	—			—			—	—	—
		4 wks.	121.59	±	10.34	127.74	±	9.79	-0.09	±	9.84	0.09	±	11.87	-0.18	-6.73 to 6.37	0.957
		8 wks.	119.95	±	9.70	126.35	±	11.88	-1.73	±	8.57	-1.30	±	11.63	-0.43	-6.56 to 5.70	0.891
		12 wks.	120.50	±	11.63	125.17	±	14.16	-1.18	±	11.67	-2.48	±	11.30	1.30	-5.61 to 8.21	0.707
Diastolic blood pressure (mmHg)		Baseline	81.41	±	8.97	84.65	±	10.10	—			—			—	—	—
		4 wks.	81.82	±	8.42	82.96	±	10.39	0.41	±	9.28	-1.70	±	7.23	2.11	-2.92 to 7.14	0.399
		8 wks.	79.23	±	10.9	81.3	±	10.92	-2.18	±	7.61	-3.35	±	8.44	1.17	-3.66 to 6.00	0.629
		12 wks.	78.77	±	10.75	81.96	±	9.21	-2.64	±	6.50	-2.70	±	8.00	0.06	-4.32 to 4.44	0.978
Values are mean ± standard deviation																	
*P < 0.05 vs. baseline within group																	
Mean difference = change from baseline (Test food group - Placebo group)																	
95% CI calculated using Welch's t method (n = 22 vs. 23)																	

Table 4: Blood test results

Items (Units)	Weeks	Measured values						Change from baseline								
														Intergroup comparison		
		Carbonated water group			Non-carbonated water group			Carbonated water group			Non-carbonated water group			Mean difference	95% CI	P-value
Triglycerides (md/dl)	Baseline	102.59	±	45.08	106.96	±	58.34	—			—			—	—	—
	4 wks.	113.68	±	56.57	116.39	±	59.49	11.09	±	31	9.43	±	42.41	1.66	-20.65 to 23.97	0.882
	8 wks.	91.68	±	40.64	129.74	±	69.37	-10.91	±	52.3	22.78	±	72.9	-33.69	-71.79 to 4.41	0.083
	12 wks.	102.45	±	48.57	127.17	±	58.88*	-0.14	±	25.7	20.22	±	45.64	-20.36	-42.65 to 1.93	0.074
Total cholesterol (md/dl)	Baseline	199.82	±	28.63	217.04	±	26.93	—			—			—	—	—
	4 wks.	202.41	±	31.82	224.61	±	39.01	2.59	±	26.28	7.57	±	23.02	-4.98	-20.57 to 10.61	0.503
	8 wks.	198	±	31.07	218.48	±	31.83	-1.82	±	30.02	1.43	±	15.41	-3.25	-16.55 to 10.05	0.653
	12 wks.	200.95	±	32.47	218.83	±	31.78	1.14	±	25.09	1.78	±	15.89	-0.64	-12.3 to 11.02	0.918
Low-density lipoprotein cholesterol (md/dl)	Baseline	118.82	±	22.39	133.74	±	23.79	—			—			—	—	—
	4 wks.	125.09	±	30.45	140.26	±	34.58	6.27	±	20.81	6.52	±	20.85	-0.25	-12.45 to 11.95	0.968
	8 wks.	120.95	±	25.99	135.04	±	30.11	2.14	±	20.03	1.30	±	14.93	0.84	-10.35 to 12.03	0.875
	12 wks.	121.64	±	28.27	134.65	±	26.77	2.82	±	17.83	0.91	±	12.15	1.91	-7.36 to 11.18	0.676
High-density lipoprotein cholesterol (md/dl)	Baseline	59.86	±	15.75	61.87	±	13.19	—			—			—	—	—
	4 wks.	58.68	±	13.12	62.22	±	15.64	-1.18	±	9.67	0.35	±	5.65	-1.53	-6.11 to 3.05	0.518
	8 wks.	59.91	±	13.26	60.57	±	13.4	0.05	±	11.2	-1.30	±	5.89	1.35	-3.49 to 6.19	0.619
	12 wks.	60.32	±	13.65	60.65	±	12.07	0.45	±	8.95	-1.22	±	5.44	1.67	-1.37 to 4.71	0.451
Phospholipids (mg/dL)	Baseline	200.68	±	25.02	214	±	20.3	—			—			—	—	—
	4 wks.	209.68	±	25.6	225.04	±	28.69*	9.00	±	24.89	11.04	±	18.1	-2.04	-15.37 to 11.29	0.754
	8 wks.	201.23	±	27.79	222.26	±	20.43*	0.55	±	29.92	8.26	±	17.2	-7.71	-22.51 to 7.09	0.299
	12 wks.	204	±	28.62	221.52	±	23.62*	3.32	±	22.06	7.52	±	14.43	-4.2	-15.66 to 7.26	0.451

Fasting blood sugar (mg/dL)		Baseline		88.73	±	6.41			91.52	±	8.8			—					—					—					—		
		4 wks.		88.55	±	9.92			90.83	±	9.6			-0.18	±	8.69			-0.70	±	6.93			0.52	-4.35	to	5.39		0.827		
		8 wks.		88.77	±	5.64			92.52	±	10.62			0.05	±	5.71			1.00	±	7.25			-0.95	-4.67	to	2.77		0.627		
		12 wks.		89.18	±	8.17			90.70	±	8.73			0.45	±	5.81			-0.83	±	5.35			1.28	-2.12	to	4.68		0.446		
Glycated hemoglobin (%)		Baseline		5.25	±	0.23			5.36	±	0.31			—					—					—					—		
		4 wks.		5.35	±	0.17*			5.43	±	0.26			0.11	±	0.13			0.07	±	0.21			0.04	-0.05	to	0.13		0.451		
		8 wks.		5.35	±	0.17*			5.42	±	0.28			0.10	±	0.17			0.06	±	0.15			0.04	-0.05	to	0.13		0.411		
		12 wks.		5.42	±	0.20*			5.47	±	0.29*			0.18	±	0.15			0.11	±	0.16			0.07	-0.02	to	0.16		0.146		
Aspartate transaminase (U/L)		Baseline		24.82	±	8.33			22.91	±	4.48			—					—					—					—		
		4 wks.		23.41	±	7.87			23.00	±	6.79			-1.41	±	7.46			0.09	±	6.34			-1.50	-5.68	to	2.68		0.472		
		8 wks.		21.55	±	5.08*			22.09	±	7.24			-3.27	±	7.26			-0.83	±	5.89			-2.44	-5.56	to	0.68		0.220		
		12 wks.		19.77	±	3.45*			21.91	±	5.62			-5.05	±	7.94			-1.00	±	3.86			-4.05	-7.57	to	-0.53		0.034		
Alanine transaminase (U/L)		Baseline		28.59	±	18.44			23.78	±	11.69			—					—					—					—		
		4 wks.		26.18	±	15.01			24.78	±	14.25			-2.41	±	16.05			1.00	±	6.30			-3.41	-10.93	to	4.11		0.349		
		8 wks.		21.59	±	9.12			25.09	±	20.52			-7.00	±	16.37			1.30	±	12.67			-8.30	-17.15	to	0.55		0.063		
		12 wks.		19.77	±	8.55*			24.43	±	15.03			-8.82	±	18.08			0.65	±	7.64			-9.47	-18.01	to	-0.93		0.026		
Gamma-glutamyl transferase (U/L)		Baseline		28.45	±	14.73			30.57	±	17.36			—					—					—					—		
		4 wks.		27.27	±	12.43			31.65	±	17.99			-1.18	±	4.49			1.09	±	6.47			-2.27	-5.77	to	1.23		0.181		
		8 wks.		25.27	±	12.26*			28.61	±	17.2			-3.18	±	6.88			-1.96	±	10.94			-1.22	-6.77	to	4.33		0.657		
		12 wks.		24.45	±	11.69*			29.83	±	16.82			-4.00	±	6.91			-0.74	±	6.80			-3.26	-7.38	to	0.86		0.118		
Alkaline phosphatase (U/L)		Baseline		61.64	±	17.38			66.30	±	16.82			—					—					—					—		
		4 wks.		61.5	±	15.78			66.13	±	17.72			-0.14	±	3.81			-0.17	±	6.18			0.03	-3.06	to	3.12		0.981		
		8 wks.		59.18	±	17			65.26	±	15.41			-2.45	±	5.62			-1.04	±	9.17			-1.41	-5.98	to	3.16		0.539		
		12 wks.		60.36	±	16.86			66.09	±	16.28			-1.27	±	9.99			-0.22	±	6.98			-1.05	-6.27	to	4.17		0.682		
Lactate dehydrogenase (U/L)		Baseline		185	±	33.04			183.48	±	24.25			—					—					—					—		
		4 wks.		180.77	±	35.37			183.65	±	23.12			-4.23	±	15.33			0.17	±	12.93			-4.40	-13.24	to	4.44		0.303		
		8 wks.		187.41	±	41.45			182.83	±	27.63			2.41	±	18.36			-0.65	±	17.83			3.06	-8.00	to	14.12		0.573		
		12 wks.		187.18	±	40.51			182.39	±	24.84			2.18	±	19.78			-1.09	±	9.63			3.27	-6.41	to	12.95		0.489		

Total bilirubin (mg/dL)	Baseline	0.73 ± 0.28	0.78 ± 0.29	—	—	—	—
	4 wks.	0.65 ± 0.32	0.76 ± 0.35	-0.07 ± 0.24	-0.03 ± 0.33	-0.04	-0.22 to 0.14
	8 wks.	0.72 ± 0.32	0.68 ± 0.22*	-0.01 ± 0.26	-0.10 ± 0.22	0.09	-0.06 to 0.24
	12 wks.	0.77 ± 0.41	0.73 ± 0.28	0.04 ± 0.31	-0.06 ± 0.28	0.10	-0.08 to 0.28
Total protein (g/dL)	Baseline	7.25 ± 0.44	7.24 ± 0.32	—	—	—	—
	4 wks.	7.2 ± 0.32	7.33 ± 0.42	-0.05 ± 0.31	0.09 ± 0.33	-0.14	-0.33 to 0.05
	8 wks.	7.12 ± 0.36	7.20 ± 0.3	-0.12 ± 0.37	-0.05 ± 0.22	-0.07	-0.26 to 0.12
	12 wks.	7.15 ± 0.35	7.20 ± 0.29	-0.09 ± 0.32	-0.04 ± 0.29	-0.05	-0.23 to 0.13
Albumin (g/dL)	Baseline	4.34 ± 0.35	4.41 ± 0.32	—	—	—	—
	4 wks.	4.35 ± 0.3	4.46 ± 0.36	0.01 ± 0.25	0.05 ± 0.2	-0.04	-0.18 to 0.10
	8 wks.	4.36 ± 0.26	4.42 ± 0.30	0.02 ± 0.3	0.01 ± 0.15	0.01	-0.14 to 0.16
	12 wks.	4.36 ± 0.3	4.39 ± 0.27	0.02 ± 0.21	-0.02 ± 0.17	0.04	-0.08 to 0.16
Uric acid (mg/dL)	Baseline	5.11 ± 1.09	5.63 ± 1.36	—	—	—	—
	4 wks.	5.33 ± 1.24	5.64 ± 1.36	0.22 ± 0.76	0.01 ± 0.64	0.21	-0.20 to 0.62
	8 wks.	5.25 ± 1.26	5.58 ± 1.4	0.14 ± 0.74	-0.05 ± 0.57	0.19	-0.19 to 0.57
	12 wks.	5.25 ± 1.23	5.52 ± 1.55	0.14 ± 0.61	-0.11 ± 0.61	0.25	-0.12 to 0.62
Urea nitrogen (mg/dL)	Baseline	13.5 ± 3.02	12.53 ± 2.62	—	—	—	—
	4 wks.	13.8 ± 3.07	12.57 ± 3.15	0.30 ± 3.07	0.03 ± 2.93	0.27	-1.53 to 2.07
	8 wks.	13.05 ± 3.07	13.22 ± 2.95	-0.46 ± 2.99	0.69 ± 2.26	-1.15	2.77 to 0.47
	12 wks.	13.37 ± 3.17	12.41 ± 3.20	-0.14 ± 2.77	-0.12 ± 2.38	-0.02	-1.57 to 1.53
Creatinine (mg/dL)	Baseline	0.78 ± 0.16	0.77 ± 0.18	—	—	—	—
	4 wks.	0.8 ± 0.18	0.77 ± 0.18	0.02 ± 0.05	0.00 ± 0.04	0.02	-0.01 to 0.05
	8 wks.	0.8 ± 0.17	0.77 ± 0.17	0.02 ± 0.04	0.00 ± 0.04	0.02	-0.01 to 0.05
	12 wks.	0.8 ± 0.18	0.76 ± 0.16	0.02 ± 0.05	-0.01 ± 0.04	0.03	0.00 to 0.06
Sodium (mEq/L)	Baseline	140.91 ± 1.41	141.26 ± 1.60	—	—	—	—
	4 wks.	140.41 ± 1.79	139.87 ± 2.10*	-0.5 ± 1.44	-1.39 ± 1.67	0.89	-0.04 to 1.82
	8 wks.	140.82 ± 1.68	140.61 ± 1.44	-0.09 ± 1.48	-0.65 ± 1.82	0.56	-0.47 to 1.59
	12 wks.	141.45 ± 2.42	141.04 ± 1.94	0.55 ± 1.63	-0.22 ± 1.62	0.77	-0.19 to 1.73
Chlorine (mEq/L)	Baseline	102.95 ± 1.76	102.96 ± 1.87	—	—	—	—
	4 wks.	103.64 ± 1.99	102.39 ± 2.02	0.68 ± 1.64	-0.57 ± 2.00	1.25	0.14 to 2.36
	8 wks.	103.77 ± 1.90*	102.65 ± 1.80	0.82 ± 1.84	-0.30 ± 1.92	1.12	-0.02 to 2.26
	12 wks.	103.86 ± 1.70*	102.61 ± 1.85	0.91 ± 1.69	-0.35 ± 2.04	1.26	0.13 to 2.39

Potassium	Baseline	4.37	±	0.42	4.24	±	0.36	—	—	—	—	—	—	—	—	—	—
(mEq/L)	4 wks.	4.82	±	0.43*	4.70	±	0.50*	0.45	±	0.43	0.46	±	0.46	-0.01	-0.27 to 0.25	0.908	
	8 wks.	4.29	±	0.4	4.11	±	0.36	-0.08	±	0.47	-0.13	±	0.30	0.05	-0.17 to 0.27	0.708	
	12 wks.	4.15	±	0.5	4.00	±	0.36*	-0.23	±	0.56	-0.23	±	0.33	0.00	-0.26 to 0.26	0.957	
Calcium	Baseline	9.23	±	0.31	9.31	±	0.35	—	—	—	—	—	—	—	—	—	
(mEq/L)	4 wks.	9.16	±	0.38	9.30	±	0.31	-0.07	±	0.27	-0.01	±	0.29	-0.06	-0.23 to 0.11	0.482	
	8 wks.	9.04	±	0.33*	9.16	±	0.27*	-0.20	±	0.25	-0.15	±	0.27	-0.05	-0.20 to 0.10	0.583	
	12 wks.	9.10	±	0.38*	9.17	±	0.27*	-0.13	±	0.28	-0.14	±	0.28	0.01	-0.16 to 0.18	0.848	
Magnesium	Baseline	2.33	±	0.13	2.39	±	0.15	—	—	—	—	—	—	—	—	—	
(mEq/L)	4 wks.	2.36	±	0.13	2.40	±	0.17	0.03	±	0.12	0.01	±	0.14	0.02	-0.05 to 0.09	0.564	
	8 wks.	2.30	±	0.15	2.31	±	0.14*	-0.04	±	0.14	-0.08	±	0.14	0.04	-0.04 to 0.12	0.280	
	12 wks.	2.39	±	0.12*	2.39	±	0.14	0.06	±	0.12	0.00	±	0.13	0.06	-0.01 to 0.13	0.094	
Serum iron	Baseline	105.14	±	47.08	120.04	±	52.74	—	—	—	—	—	—	—	—	—	
(µg/dL)	4 wks.	93.73	±	35.56	119.17	±	35.04	-11.41	±	44.09	-0.87	±	58.79	-10.54	-41.28 to 20.20	0.501	
	8 wks.	102.27	±	45.63	105.09	±	25.44	-2.86	±	59.06	-14.96	±	62.14	12.10	-24.45 to 48.65	0.507	
	12 wks.	105.73	±	39.4	110.43	±	39.86	0.59	±	44.38	-9.61	±	49.53	10.20	-17.55 to 37.95	0.472	
Values are mean ± standard deviation																	
*P < 0.05 vs. baseline within group																	
Mean difference = change from baseline (Test food group - Placebo group)																	
95% CI calculated using Welch's t method (n = 22 vs. 23)																	

Items	Weeks	Measured values		Change from baseline					Intergroup comparison			
		Carbonated water group	Non-carbonated water group	Carbonated water group	Non-carbonated water group	Mean difference	95%CI					P-value
Calorific value	Baseline	1331 ± 2.2	2053.8	12493.4 ±	2470.1	—	—	—	—	—	—	—

(kcal/week)	4 wks.	1337 ± 8.1	2621.3	12466.6 ± 2639.3	65.9 ± 1	2162.1	-26.9 ± 1722.8	92.8	1088.21	- t o	1273.81	0.874
	8 wks.	1263 ± 6.8	1776	12054 ± 2462.9	-675.5 ± 7	1534.7	-439.5 ± 1781.4	-236.0	1234.58	- t o	762.58	0.637
	12 wks.	1245 ± 0.3	1680.8*	12307.6 ± 2347.9	-861.9 ± 6	1775.6	-185.8 ± 1931.4	-676.1	1790.80	- t o	438.60	0.229
Protein	Basel ine	402.6 ± 77.8		387.4 ± 79.5	—	—	—	—	—	—	—	—
(g/week)	4 wks.	411.2 ± 97.7		383.4 ± 72.2	8.6 ± 88.8	88.8	-4.0 ± 64.1	12.60	34.31	- t o	59.51	0.586
	8 wks.	393.8 ± 62.9		374.2 ± 79.9	-8.8 ± 69.9	69.9	-13.2 ± 72.7	4.40	38.47	- t o	47.27	0.835
	12 wks.	385.1 ± 63.1		385.0 ± 92.1	-17.5 ± 81.2	81.2	-2.3 ± 82.6	-15.20	64.45	- t o	34.05	0.539
Carbohydrates	Basel ine	1864.6 ± 377.8		1806.9 ± 380.5	—	—	—	—	—	—	—	—
(g/week)	4 wks.	1826.0 ± 389.7		1761.5 ± 366	-38.6 ± 271.1	271.1	-45.4 ± 227.6	6.80	144.23	- t o	157.83	0.927
	8 wks.	1719.1 ± 310.0*		1664.2 ± 355.2*	-145.5 ± 231.7	231.7	-142.7 ± 261.6	-2.80	151.23	- t o	145.63	0.970

	12 wks.	1669.6 ± 332.1 *	1740.1 ± 303.8	-195.1 ± 258	-66.8 ± 248.1	-128.3	-280.61	-	t o	24.01	0.096
Lipids	Baseline	431.3 ± 81.4	382.7 ± 117.0	—	—	—	—	—	—	—	—
(g/week)	4 wks.	444.9 ± 131.6	389.6 ± 105.9	13.6 ± 129.6	6.8 ± 69.5	6.80	56.77	-	t o	70.37	0.830
	8 wks.	422.1 ± 71.7	396.3 ± 117.5	-9.2 ± 77.6	13.6 ± 93.1	-22.80	74.27	-	t o	28.67	0.378
	12 wks.	431.0 ± 90.2	388.9 ± 109.3	-0.3 ± 111.4	6.2 ± 102.3	-6.50	70.91	-	t o	57.91	0.839
Snack only: calorific value	Baseline	1708.3 ± 1153.5	1617 ± 1155.0	—	—	—	—	—	—	—	—
(kcal/week)	4 wks.	2076.8 ± 1734.7	1745 ± 1246.0	368.5 ± 1090.4	128.0 ± 858.3	240.5	352.50	-	t o	833.50	0.414
	8 wks.	1538.2 ± 1317.7	1456.6 ± 1046.2	-170.1 ± 764.8	-160.4 ± 690.5	-9.70	448.66	-	t o	429.26	0.964
	12 wks.	1434.3 ± 1136.9	1597.7 ± 907.8	-274.0 ± 867.9	-19.3 ± 666.3	-254.70	722.58	-	t o	213.18	0.274
Snacks only: Protein	Baseline	29.9 ± 25.9	28.4 ± 21.6	—	—	—	—	—	—	—	—
(g/week)	4 wks.	35.4 ± 28.9	35.7 ± 28.4	5.5 ± 16.3	7.3 ± 20.1	-1.80	12.79	-	t o	9.19	0.733

	8 wks.	24.0 ± 20.3	23.9 ± 19.1	-5.9 ± 17.3	-4.5 ± 16.8	-1.40	- t 11.66 o	8.86	0.78 2
	12 wks.	23.3 ± 20.7	25.4 ± 16.7	-6.7 ± 17.5	-3.1 ± 14.4	-3.60	- t 13.27 o	6.07	0.45 3
Snacks only: Carbohydrates (g/week)	Baseline	245.2 ± 159.7	236.6 ± 191.5	—	—	—	— — —	—	—
	4 wks.	284.3 ± 259.7	237.6 ± 183.5	39.1 ± 158.6	1.0 ± 111	38.1	- t 44.89 o	121.09	0.35 4
	8 wks.	198.0 ± 175.4	207.9 ± 187.4	-47.1 ± 122.7	-28.6 ± 114.3	-18.5	- t 89.90 o	52.9	0.60 3
	12 wks.	179.9 ± 149.7 *	232.9 ± 166.1	-65.3 ± 124.3	-3.7 ± 93.9	-61.6	- t 128.2 o 4	5.04	0.06 7
Snack only: Lipids (g/week)	Baseline	57.5 ± 50.8	51.1 ± 40.4	—	—	—	— — —	—	—
	4 wks.	75.5 ± 70.5	58.5 ± 44.3	18.0 ± 51.0	7.4 ± 34.2	10.60	- t 15.76 o	36.96	0.41 6
	8 wks.	54.8 ± 65.5	47.0 ± 32.9	-2.7 ± 35.7	-4.1 ± 27.6	1.40	- t 17.89 o	20.69	0.88 5
	12 wks.	58.6 ± 55.1	52.0 ± 33.7	1.1 ± 34.9	0.9 ± 33.6	0.20	- t 20.41 o	20.81	0.98 4

Values are mean ± standard

deviation

*P < 0.05 vs. baseline within

group

Mean difference = change from baseline (Carbonated water group – Non-

carbonated water group)

95% CI calculated using Welch's t

method (n = 22 vs. 23)

Items (Units)	Weeks	Measured values		Change from baseline		Intergroup comparison		
		Carbonated water group	Non-carbonated water group	Carbonated water group	Non-carbonated water group	Mean difference	95% CI	P-value
Body weight	Baseline	68.57 ± 9.10	68.83 ± 9.62	—	—	—	—	—
(kg)	4 wks.	69.30 ± 9.07*	69.10 ± 9.63	0.73 ± 1.08	0.28 ± 1.23	0.45	-0.22 to 1.12	0.201
	8 wks.	68.85 ± 9.01	69.22 ± 9.35	0.28 ± 1.15	0.40 ± 1.12	-0.12	-0.80 to 0.56	0.728

	12 wks.	68.45 ± 9.27	69.42 ± 9.44*	-0.12 ± 1.25	0.60 ± 0.96	-0.72	-1.37	-0.07	0.036
Body mass index (kg/m ²)	Baseline	25.26 ± 1.49	25.33 ± 1.38	—	—	—	—	—	—
	4 wks.	25.55 ± 1.52*	25.43 ± 1.32	0.29 ± 0.42	0.10 ± 0.43	0.19	0.05	0.43	0.134
	8 wks.	25.39 ± 1.57	25.48 ± 1.28	0.12 ± 0.43	0.15 ± 0.38	-0.03	0.26	0.20	0.836
	12 wks.	25.22 ± 1.63	25.55 ± 1.32*	-0.04 ± 0.46	0.22 ± 0.36	-0.26	0.50	0.02	0.037
Systolic blood pressure (mmHg)	Baseline	121.68 ± 10.65	127.65 ± 12.12	—	—	—	—	—	—
	4 wks.	121.59 ± 10.34	127.74 ± 9.79	-0.09 ± 9.84	0.09 ± 11.87	-0.18	6.73	6.37	0.957
	8 wks.	119.95 ± 9.70	126.35 ± 11.88	-1.73 ± 8.57	-1.30 ± 11.63	-0.43	6.56	5.70	0.891
	12 wks.	120.50 ± 11.63	125.17 ± 14.16	-1.18 ± 11.67	-2.48 ± 11.30	1.30	5.61	8.21	0.707

Diastolic blood pressure (mmHg)	Baseline	Carbonated water group			Non-carbonated water group			Carbonated water group			Non-carbonated water group		
		Mean	±	SD	Mean	±	SD	Mean difference	95% CI	P-value	Mean difference	95% CI	P-value
		81.41	±	8.97	84.65	±	10.10	—	—	—	—	—	—
	4 wks.	81.82	±	8.42	82.96	±	10.39	0.41	± 9.28	—	−1.70	± 7.23	0.39
	8 wks.	79.23	±	10.9	81.3	±	10.92	−2.18	± 7.61	—	−3.35	± 8.44	0.62
	12 wks.	78.77	±	10.75	81.96	±	9.21	−2.64	± 6.50	—	−2.70	± 8.00	0.97

Values are mean ± standard

deviation

*P < 0.05 vs. baseline within group

Mean difference = change from baseline (Carbonated water group – Non-

carbonated water group)

95% CI calculated using Welch's t method (n =

22 vs. 23)

Items	Weeks	Measured values				Change from baseline						
		Carbonated water group		Non-carbonated water group		Carbonated water group		Non-carbonated water group		Intergroup comparison		
(Units)										Mean difference	95% CI	P-value
Frequency of snacks	Baseline	8.07	± 6.23	7.50	± 5.79	—		—		—	—	—
(times/week)	4 wks.	6.80	± 5.31*	7.62	± 5.91	-1.27	± 1.88	0.12	± 1.93	-1.39	-2.82 to 0.28	0.018
	8 wks.	6.75	± 5.53*	7.63	± 5.72	-1.32	± 2.63	0.13	± 2.25	-1.45	-3.01 to 0.11	0.053
	12 wks.	6.82	± 5.38*	7.74	± 5.98	-1.25	± 2.49	0.24	± 2.44	-1.49	-2.93 to -0.05	0.049
Amount of alcohol consumed: Alcohol equivalent	Baseline	68.00	± 48.82	56.93	± 56.36	—		—		—	—	—
(g/week)	4 wks.	65.45	± 48.67	60.99	± 52.11	-2.55	± 21.15	4.06	± 28.17	-6.61	-22.96 to 9.74	0.380
	8 wks.	63.00	± 47.04	64.23	± 49.30	-5.00	± 25.68	7.31	± 37.65	-12.31	-33.41 to 8.79	0.209

12 wks.	52.6 4	±	42.66 *	66.01	±	55.31	-15.3 6	±	30.7 9	9.08	±	36.12	-24.44	-44.02	t o	-4.86	0.01 9
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Values are mean ± standard

deviation

*P < 0.05 vs. baseline within

group

Mean difference = change from baseline (Carbonated water group – Non-

carbonated water group)

95% CI calculated using Welch's t method (n =

22 vs. 23)

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Items (Units)	Week s	Measured values		Change from baseline				
		Carbonated water group	Non-carbonated water group	Carbonated water group	Non-carbonated water group	Mean difference	95% CI	P- valu e

Triglycerides	Basel ine	102.59	± 45.08	106.96	± 58.34	—		—		—		—		—	
(md/dl)	4 wks.	113.68	± 56.57	116.39	± 59.49	11.09	± 31	9.43	± 42.41	1.66	-20.65	t o	23.97	0.882	
	8 wks.	91.68	± 40.64	129.74	± 69.37	-10.91	± 52.3	22.78	± 72.9	-33.69	-71.79	t o	4.41	0.083	
	12 wks.	102.45	± 48.57	127.17	± 58.88*	-0.14	± 25.7	20.22	± 45.64	-20.36	-42.65	t o	1.93	0.074	
Total cholesterol	Basel ine	199.82	± 28.63	217.04	± 26.93	—		—		—		—		—	
(md/dl)	4 wks.	202.41	± 31.82	224.61	± 39.01	2.59	± 26.28	7.57	± 23.02	-4.98	-20.57	t o	10.61	0.503	
	8 wks.	198	± 31.07	218.48	± 31.83	-1.82	± 30.02	1.43	± 15.41	-3.25	-16.55	t o	10.05	0.653	
	12 wks.	200.95	± 32.47	218.83	± 31.78	1.14	± 25.09	1.78	± 15.89	-0.64	-12.3	t o	11.02	0.918	
Low-density lipoprotein cholesterol	Basel ine	118.82	± 22.39	133.74	± 23.79	—		—		—		—		—	

(md/dl)	4 wks.	125.09	± 30.45	140.26	± 34.58	6.27	± 20.81	6.52	± 20.85	-0.25	12.45	- t o	11.95	0.968
	8 wks.	120.95	± 25.99	135.04	± 30.11	2.14	± 20.03	1.30	± 14.93	0.84	10.35	- t o	12.03	0.875
	12 wks.	121.64	± 28.27	134.65	± 26.77	2.82	± 17.83	0.91	± 12.15	1.91	7.36	- t o	11.18	0.676
High-density lipoprotein cholesterol	Baseline	59.86	± 15.75	61.87	± 13.19	—	—	—	—	—	—	—	—	—
(md/dl)	4 wks.	58.68	± 13.12	62.22	± 15.64	-1.18	± 9.67	0.35	± 5.65	-1.53	6.11	- t o	3.05	0.518
	8 wks.	59.91	± 13.26	60.57	± 13.4	0.05	± 11.2	-1.30	± 5.89	1.35	3.49	- t o	6.19	0.619
	12 wks.	60.32	± 13.65	60.65	± 12.07	0.45	± 8.95	-1.22	± 5.44	1.67	1.37	- t o	4.71	0.451
Phospholipids	Baseline	200.68	± 25.02	214	± 20.3	—	—	—	—	—	—	—	—	—
(mg/dL)	4 wks.	209.68	± 25.6	225.04	± 28.69*	9.00	± 24.89	11.04	± 18.1	-2.04	15.37	- t o	11.29	0.754
	8 wks.	201.23	± 27.79	222.26	± 20.43*	0.55	± 29.92	8.26	± 17.2	-7.71	22.51	- t o	7.09	0.299

	12 wks.	204	±	28.62	221.52	±	23.62*	3.32	±	22.0 6	7.52	±	14.43	-4.2	15.6 6	t o	7.2 6	0.45 1
Fasting blood sugar (mg/dL)	Basel ine 4 wks. 8 wks. 12 wks.	88.73	±	6.41	91.52	±	8.8	—			—			—		—		—
		88.55	±	9.92	90.83	±	9.6	-0.18	±	8.69	-0.70	±	6.93	0.52	- 4.35	t o	5.3 9	0.82 7
		88.77	±	5.64	92.52	±	10.62	0.05	±	5.71	1.00	±	7.25	-0.95	- 4.67	t o	2.7 7	0.62 7
		89.18	±	8.17	90.70	±	8.73	0.45	±	5.81	-0.83	±	5.35	1.28	- 2.12	t o	4.6 8	0.44 6
Glycated hemoglobin (%)	Basel ine 4 wks. 8 wks. 12 wks.	5.25	±	0.23	5.36	±	0.31	—			—			—		—		—
		5.35	±	0.17*	5.43	±	0.26	0.11	±	0.13	0.07	±	0.21	0.04	- 0.05	t o	0.1 3	0.45 1
		5.35	±	0.17*	5.42	±	0.28	0.10	±	0.17	0.06	±	0.15	0.04	- 0.05	t o	0.1 3	0.41 1
		5.42	±	0.20*	5.47	±	0.29*	0.18	±	0.15	0.11	±	0.16	0.07	- 0.02	t o	0.1 6	0.14 6
Aspartate transaminase (U/L)	Basel ine 4 wks. 8 wks.	24.82	±	8.33	22.91	±	4.48	—			—			—		—		—
		23.41	±	7.87	23.00	±	6.79	-1.41	±	7.46	0.09	±	6.34	-1.50	- 5.68	t o	2.6 8	0.47 2
		21.55	±	5.08*	22.09	±	7.24	-3.27	±	7.26	-0.83	±	5.89	-2.44	- 5.56	t o	0.6 8	0.22 0

	12 wks.	19.77 ± 3.45*	21.91 ± 5.62	-5.05 ± 7.94	-1.00 ± 3.86	-4.05	-7.57	-0.53	0.034
Alanine transaminase	Baseline	28.59 ± 18.44	23.78 ± 11.69	—	—	—	—	—	—
(U/L)	4 wks.	26.18 ± 15.01	24.78 ± 14.25	-2.41 ± 16.05	1.00 ± 6.30	-3.41	10.93	-4.11	0.349
	8 wks.	21.59 ± 9.12	25.09 ± 20.52	-7.00 ± 16.37	1.30 ± 12.67	-8.30	17.15	-0.55	0.063
	12 wks.	19.77 ± 8.55*	24.43 ± 15.03	-8.82 ± 18.08	0.65 ± 7.64	-9.47	18.01	-0.93	0.026
Gamma-glutamyl transferase	Baseline	28.45 ± 14.73	30.57 ± 17.36	—	—	—	—	—	—
(U/L)	4 wks.	27.27 ± 12.43	31.65 ± 17.99	-1.18 ± 4.49	1.09 ± 6.47	-2.27	5.77	-1.23	0.181
	8 wks.	25.27 ± 12.26*	28.61 ± 17.2	-3.18 ± 6.88	-1.96 ± 10.94	-1.22	6.77	-4.33	0.657
	12 wks.	24.45 ± 11.69*	29.83 ± 16.82	-4.00 ± 6.91	-0.74 ± 6.80	-3.26	7.38	-0.86	0.118
Alkaline phosphatase	Baseline	61.64 ± 17.38	66.30 ± 16.82	—	—	—	—	—	—

(U/L)	4 wks.	61.5 ± 15.78	66.13 ± 17.72	-0.14 ± 3.81	-0.17 ± 6.18	0.03	- t 3.1 3.06 o 2	0.98
	8 wks.	59.18 ± 17	65.26 ± 15.41	-2.45 ± 5.62	-1.04 ± 9.17	-1.41	- t 3.1 5.98 o 6	0.53
	12 wks.	60.36 ± 16.86	66.09 ± 16.28	-1.27 ± 9.99	-0.22 ± 6.98	-1.05	- t 4.1 6.27 o 7	0.68
Lactate dehydrogenase	Baseline	185 ± 33.04	183.48 ± 24.25	—	—	—	—	—
(U/L)	4 wks.	180.7 ± 35.37	183.65 ± 23.12	-4.23 ± 15.3 3	0.17 ± 12.93	-4.40	- t 4.4 13.2 o 4	0.30
	8 wks.	187.4 ± 41.45	182.83 ± 27.63	2.41 ± 18.3 6	-0.65 ± 17.83	3.06	- t 14. 8.00 o 12	0.57
	12 wks.	187.1 ± 40.51	182.39 ± 24.84	2.18 ± 19.7 8	-1.09 ± 9.63	3.27	- t 12. 6.41 o 95	0.48
Total bilirubin	Baseline	0.73 ± 0.28	0.78 ± 0.29	—	—	—	—	—
(mg/dL)	4 wks.	0.65 ± 0.32	0.76 ± 0.35	-0.07 ± 0.24	-0.03 ± 0.33	-0.04	- t 0.1 0.22 o 4	0.59
	8 wks.	0.72 ± 0.32	0.68 ± 0.22*	-0.01 ± 0.26	-0.10 ± 0.22	0.09	- t 0.2 0.06 o 4	0.21
	12 wks.	0.77 ± 0.41	0.73 ± 0.28	0.04 ± 0.31	-0.06 ± 0.28	0.10	- t 0.2 0.08 o 8	0.27
Total protein	Baseline	7.25 ± 0.44	7.24 ± 0.32	—	—	—	—	—

(g/dL)	4 wks.	7.2 ± 0.32	7.33 ± 0.42	-0.05 ± 0.31	0.09 ± 0.33	-0.14	- t 0.0	0.17
							0.33 o 5	5
	8 wks.	7.12 ± 0.36	7.20 ± 0.3	-0.12 ± 0.37	-0.05 ± 0.22	-0.07	- t 0.1	0.41
							0.26 o 2	1
	12 wks.	7.15 ± 0.35	7.20 ± 0.29	-0.09 ± 0.32	-0.04 ± 0.29	-0.05	- t 0.1	0.60
							0.23 o 3	4
Albumin	Baseline	4.34 ± 0.35	4.41 ± 0.32	—	—	—	—	—
(g/dL)	4 wks.	4.35 ± 0.3	4.46 ± 0.36	0.01 ± 0.25	0.05 ± 0.2	-0.04	- t 0.1	0.52
							0.18 o 0	4
	8 wks.	4.36 ± 0.26	4.42 ± 0.30	0.02 ± 0.3	0.01 ± 0.15	0.01	- t 0.1	0.94
							0.14 o 6	2
	12 wks.	4.36 ± 0.3	4.39 ± 0.27	0.02 ± 0.21	-0.02 ± 0.17	0.04	- t 0.1	0.43
							0.08 o 6	5
Uric acid	Baseline	5.11 ± 1.09	5.63 ± 1.36	—	—	—	—	—
(mg/dL)	4 wks.	5.33 ± 1.24	5.64 ± 1.36	0.22 ± 0.76	0.01 ± 0.64	0.21	- t 0.6	0.32
							0.20 o 2	1
	8 wks.	5.25 ± 1.26	5.58 ± 1.4	0.14 ± 0.74	-0.05 ± 0.57	0.19	- t 0.5	0.35
							0.19 o 7	5
	12 wks.	5.25 ± 1.23	5.52 ± 1.55	0.14 ± 0.61	-0.11 ± 0.61	0.25	- t 0.6	0.17
							0.12 o 2	9
Urea nitrogen	Baseline	13.5 ± 3.02	12.53 ± 2.62	—	—	—	—	—
(mg/dL)	4 wks.	13.8 ± 3.07	12.57 ± 3.15	0.30 ± 3.07	0.03 ± 2.93	0.27	- t 2.0	0.77
							1.53 o 7	2

	8 wks.	13.05 ± 3.07	13.22 ± 2.95	-0.46 ± 2.99	0.69 ± 2.26	-1.15	2.77	t o 7	0.4	0.15
	12 wks.	13.37 ± 3.17	12.41 ± 3.20	-0.14 ± 2.77	-0.12 ± 2.38	-0.02	1.57	- t o 3	1.5	0.98
Creatinine	Baseline	0.78 ± 0.16	0.77 ± 0.18	—	—	—	—	—	—	—
(mg/dL)	4 wks.	0.8 ± 0.18	0.77 ± 0.18	0.02 ± 0.05	0.00 ± 0.04	0.02	0.01	- t o 5	0.0	0.14
	8 wks.	0.8 ± 0.17	0.77 ± 0.17	0.02 ± 0.04	0.00 ± 0.04	0.02	0.01	- t o 5	0.0	0.25
	12 wks.	0.8 ± 0.18	0.76 ± 0.16	0.02 ± 0.05	-0.01 ± 0.04	0.03	0.00	t o 6	0.0	0.02
Sodium	Baseline	140.9 ± 1.41	141.26 ± 1.60	—	—	—	—	—	—	—
(mEq/L)	4 wks.	140.4 ± 1.79	139.87 ± 2.10*	-0.5 ± 1.44	-1.39 ± 1.67	0.89	0.04	- t o 2	1.8	0.06
	8 wks.	140.8 ± 1.68	140.61 ± 1.44	-0.09 ± 1.48	-0.65 ± 1.82	0.56	0.47	- t o 9	1.5	0.26
	12 wks.	141.4 ± 2.42	141.04 ± 1.94	0.55 ± 1.63	-0.22 ± 1.62	0.77	0.19	- t o 3	1.7	0.12
Chlorine	Baseline	102.9 ± 1.76	102.96 ± 1.87	—	—	—	—	—	—	—
(mEq/L)	4 wks.	103.6 ± 1.99	102.39 ± 2.02	0.68 ± 1.64	-0.57 ± 2.00	1.25	0.14	t o 6	2.3	0.02
	8 wks.	103.7 ± 1.90*	102.65 ± 1.80	0.82 ± 1.84	-0.30 ± 1.92	1.12	0.02	- t o 6	2.2	0.05

	12 wks.	103.8 6	± 1.70*	102.61	± 1.85	0.91	± 1.69	-0.35	± 2.04	1.26	0.13	t o	2.3 9	0.03 0
Potassium	Basel ine	4.37	± 0.42	4.24	± 0.36	—		—		—		—		—
(mEq/L)	4 wks.	4.82	± 0.43*	4.70	± 0.50*	0.45	± 0.43	0.46	± 0.46	-0.01	-	t o	0.2 5	0.90 8
	8 wks.	4.29	± 0.4	4.11	± 0.36	-0.08	± 0.47	-0.13	± 0.30	0.05	-	t o	0.2 7	0.70 8
	12 wks.	4.15	± 0.5	4.00	± 0.36*	-0.23	± 0.56	-0.23	± 0.33	0.00	-	t o	0.2 6	0.95 7
Calcium	Basel ine	9.23	± 0.31	9.31	± 0.35	—		—		—		—		—
(mEq/L)	4 wks.	9.16	± 0.38	9.30	± 0.31	-0.07	± 0.27	-0.01	± 0.29	-0.06	-	t o	0.1 1	0.48 2
	8 wks.	9.04	± 0.33*	9.16	± 0.27*	-0.20	± 0.25	-0.15	± 0.27	-0.05	-	t o	0.1 0	0.58 3
	12 wks.	9.10	± 0.38*	9.17	± 0.27*	-0.13	± 0.28	-0.14	± 0.28	0.01	-	t o	0.1 8	0.84 8
Magnesium	Basel ine	2.33	± 0.13	2.39	± 0.15	—		—		—		—		—
(mEq/L)	4 wks.	2.36	± 0.13	2.40	± 0.17	0.03	± 0.12	0.01	± 0.14	0.02	-	t o	0.0 9	0.56 4
	8 wks.	2.30	± 0.15	2.31	± 0.14*	-0.04	± 0.14	-0.08	± 0.14	0.04	-	t o	0.1 2	0.28 0
	12 wks.	2.39	± 0.12*	2.39	± 0.14	0.06	± 0.12	0.00	± 0.13	0.06	-	t o	0.1 3	0.09 4

Serum iron	Baseline	105.14	± 47.08	120.04	± 52.74	—		—		—	—		—	
(µg/dL)	4 wks.	93.73	± 35.56	119.17	± 35.04	-11.41	± 44.09	-0.87	± 58.79	-10.54	-41.28	t o 20	20.1	0.50
	8 wks.	102.27	± 45.63	105.09	± 25.44	-2.86	± 59.06	-14.96	± 62.14	12.10	-24.45	t o 65	48.7	0.50
	12 wks.	105.73	± 39.4	110.43	± 39.86	0.59	± 44.38	-9.61	± 49.53	10.20	-17.55	t o 95	37.2	0.47

Values are mean ± standard

deviation

*P < 0.05 vs. baseline within group

Mean difference = change from baseline (Carbonated water group – Non-carbonated water group)

95% CI calculated using Welch's t method (n =

22 vs. 23)

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